

Summary and Recommendation

1. Project Objective

The objective of this project was to perform an Exploratory Data Analysis (EDA) on the Netflix Movies and TV Shows dataset to uncover key insights regarding content distribution, release trends, ratings, genres, production countries, and duration patterns. With the rapid growth of streaming platforms, understanding the composition of Netflix's content library provides valuable insights into its content strategy, audience targeting, and global expansion efforts.

The analysis aimed to answer important business questions such as:

- What type of content dominates Netflix's catalog?
- How has Netflix's content library evolved over time?
- Which countries contribute the most content?
- What genres are most popular on the platform?
- What audience segments are primarily targeted?
- How do movie durations and TV show seasons vary across the catalog?

By analyzing these factors, the project seeks to provide a comprehensive understanding of Netflix's content ecosystem and identify trends that characterize the platform's success.

2. Data Understanding and Preparation

The dataset contains detailed information about Netflix titles, including:

- Show ID
- Title
- Type (Movie or TV Show)
- Director
- Cast
- Country
- Date Added
- Release Year
- Rating
- Duration
- Listed Categories (Genres)
- Description

Data Cleaning Performed

To ensure accurate analysis, several preprocessing steps were carried out:

- Examined dataset dimensions, column names, and data types.
- Identified missing values across multiple columns.
- Handled missing values in Country, Director, Cast, Rating, and Date Added.
- Converted the Date Added column into datetime format for time-based analysis.
- Extracted year information where necessary.
- Standardized duration values for movies and TV shows.
- Checked for duplicate records and verified data consistency.
- Performed exploratory checks to identify anomalies and ensure data quality.

The dataset was successfully cleaned and transformed, making it suitable for visualization and exploratory analysis.

3. Exploratory Data Analysis (EDA)

Content Type Distribution

The first analysis focused on understanding the composition of Netflix's content library.

Findings

- Movies account for the majority of content available on Netflix.
- TV Shows represent a smaller but significant portion of the catalog.
- The difference between Movies and TV Shows indicates Netflix's strong emphasis on film-based content.

Business Insight

A movie-dominated catalog enables Netflix to appeal to a broader audience through shorter viewing commitments while continuing to expand its TV show offerings to increase long-term viewer engagement.

Content Growth Over Time

The release year analysis was conducted to understand how Netflix's content library has evolved over time.

Findings

- Content production remained relatively low before 2000.
- A significant increase occurred after 2000.
- The most rapid growth was observed between 2015 and 2019.
- Movie releases consistently exceeded TV Show releases across most years.
- Netflix experienced a substantial expansion in content volume during the streaming boom.

Business Insight

The sharp increase in content production reflects Netflix's aggressive investment in original content and global expansion strategies. A larger content library helps attract and retain subscribers by providing greater viewing variety.

Country-Wise Content Analysis

The distribution of content by country was examined to identify major contributors to Netflix's catalog.

Findings

- The United States contributes the highest number of titles.
- Several international countries also make significant contributions.
- Netflix's catalog contains content from diverse geographical regions.
- International productions have grown considerably over time.

Business Insight

The dominance of U.S. content highlights Netflix's strong roots in Hollywood, while the growing share of international content demonstrates its strategy of serving global audiences through localization and regional storytelling.

Genre and Category Analysis

The category distribution was analyzed to determine which types of content are most common.

Findings

The most frequently occurring categories include:

- Dramas
- International Movies

- Documentaries
- Stand-Up Comedy
- Romantic Movies and Shows

Drama emerged as the most dominant category, followed closely by international content.

Business Insight

The popularity of drama and international content suggests that Netflix focuses on genres with broad audience appeal. Investment in diverse genres helps the platform cater to varying viewer preferences across different markets.

Audience Rating Analysis

Ratings were analyzed to understand the primary audience targeted by Netflix content.

Findings

- TV-MA is the most common content rating.
- TV-14 ranks as the second most frequent rating.
- Family and children's content represents a smaller share of the catalog.
- Mature audience content dominates the platform.

Business Insight

Netflix primarily targets adult and teenage audiences through mature storytelling and diverse content themes. This aligns with the viewing behavior of its largest subscriber groups.

Duration Analysis

The duration of both Movies and TV Shows was explored to understand content structure.

Movie Duration Findings

- Most movies range between 80 and 120 minutes.
- Extremely short and extremely long movies are relatively rare.
- The duration distribution follows standard feature-film industry practices.

TV Show Duration Findings

- Most TV Shows consist of a single season.

- The number of shows decreases as the season count increases.
- Multi-season series represent a smaller portion of the overall catalog.

Business Insight

The prevalence of standard-length movies and single-season TV Shows reflects modern streaming trends where viewers often prefer concise and easily consumable content.

4. Major Insights from the Analysis

Based on the exploratory analysis, several key insights emerged:

- Movies dominate Netflix's content library.
- Content production increased dramatically after 2000.
- The United States is the largest contributor of Netflix titles.
- Drama is the most common genre on the platform.
- International Movies represent a significant portion of the catalog.
- TV-MA is the most prevalent audience rating.
- Most movies have runtimes between 80 and 120 minutes.
- Single-season TV Shows are more common than long-running series.
- Netflix has steadily diversified its catalog through international expansion and genre variety.

These patterns collectively illustrate Netflix's strategy of balancing global content diversity with mass-market entertainment.

5. Business Recommendations

Recommendation 1: Continue Investing in International Content

The strong presence and popularity of international titles suggest growing demand for localized content. Netflix should continue expanding regional productions to strengthen its global market position.

Recommendation 2: Expand High-Performing Genres

Dramas, documentaries, and international movies consistently appear among the most popular categories. Increasing investments in these genres may improve audience engagement.

Recommendation 3: Strengthen Original Content Strategy

The rapid growth observed after 2015 demonstrates the effectiveness of Netflix's original content investments. Continued focus on original productions can help maintain competitive differentiation.

Recommendation 4: Diversify Family and Children's Content

Since mature content dominates the catalog, increasing family-friendly offerings could attract additional audience segments and improve household subscriptions.

Recommendation 5: Leverage Regional Viewing Preferences

Country-level content trends can be used to develop targeted content acquisition and production strategies for different markets.

Recommendation 6: Focus on Viewer Retention Through Series Content

Although movies dominate the catalog, successful multi-season TV Shows often generate long-term engagement. Increasing investments in high-quality episodic content can improve subscriber retention.

Conclusion

This analysis provided valuable insights into Netflix's content library, highlighting key trends in content distribution, release patterns, genres, ratings, and global production. The findings show that Netflix is primarily movie-focused, with strong contributions from the United States and a growing emphasis on international content. Popular genres such as Drama and International Movies, along with the dominance of TV-MA rated content, reflect the platform's strategy of targeting diverse and mature audiences. Overall, Netflix's continuous content expansion and genre diversification have played a significant role in establishing it as one of the world's leading streaming platforms.